Climate Policy Impact Analysis

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1 Construction of the Policy Index

To quantify the strength of climate-related policy portfolios across countries and years, we constructed the Composite Policy Strength Index (CPSI) from the IEA policy dataset. The process followed these main steps:

1.1 Data Preparation and Expansion

We began with a cleaned version of the IEA climate policy dataset. Each policy was filtered and classified based on its implementation status. Policies marked as `in force` were temporally expanded: for each such policy, yearly duplicates were created for all years from the initial implementation year until the last available year in the dataset. Policies with `ended` status were retained only in their original year.

1.2 Policy Scoring via Text Classification

To score the strength of individual policies, we used a rule-based Natural Language Processing (NLP) approach. A set of keywords was defined to map policy content to one of four conceptual levels of intervention, inspired by regulatory and economic theory:

- **C1 (1.00):** Strong mandates or bans (e.g., "prohibit", "ban", "phase-out")
- **C2 (0.75):** Market-based instruments (e.g., "tax", "pricing", "levy")
- **C3 (0.50):** Standards and compliance mechanisms (e.g., "emissions limit", "performance")
- **C4 (0.25):** Informational and financial support (e.g., "grant", "education", "funding")

Each policy's text (including topic and description fields) was scanned for these keywords and assigned a numeric score corresponding to its highest-matching intervention level.

1.3 Index Aggregation

For each country c and year y, we computed the following summary metrics:

- $\bar{s}_{c,y}$: The average policy score (across all policies for that country-year)
- $n_{c,y}$: The number of policies implemented
- $\ln(1 + n_{c,y})$: A log-transformed count of policies to impose diminishing returns
- $\text{CPSI}_{c,y} = \bar{s}_{c,y} \cdot \ln(1 + n_{c,y})$: The composite index

This formulation rewards both policy ambition and policy proliferation, while attenuating the effect of sheer quantity through logarithmic scaling. The resulting CPSI serves as a synthetic indicator of climate policy strength over time and across countries.

2 Alternative CPSI Formulations

While the initial formulation of the CPSI—based on the product of average policy score and the logarithm of policy count—provided a simple and interpretable metric, it exhibited some limitations. In particular, the logarithmic transformation, while useful for imposing diminishing returns, disproportionately dampened the contribution of countries with a large number of high-scoring policies. This compression effect reduced the ability of the index to distinguish between genuinely ambitious policy portfolios. To address this, we explored alternative formulations that allow for more flexible scaling of policy count and quality.

2.1 Power–Transformed CPSI

The power (Box–Cox) variant retains diminishing returns while providing a steeper reward for additional policies:

$$\text{CPSI}_{c,y}^{\text{power}} = \bar{s}_{c,y} \left(\frac{(1 + n_{c,y})^{\theta} - 1}{\theta} \right) \qquad 0 < \theta < 1.$$

1. $\bar{s}_{c,y}$, $n_{c,y}$: defined as above.
2. θ: curvature parameter. $\theta \to 0$ reproduces the natural logarithm; larger θ values approach a linear scale, increasing the dispersion between low and high policy counts while still imposing diminishing returns.

2.2 Geometric–Mean CPSI

This specification treats policy *quality* and *quantity* as separate elasticities combined multiplicatively:

$$\mathrm{CPSI}_{c,y}^{\mathrm{geom}} = (\bar{s}_{c,y})^{\beta}\,(1 + n_{c,y})^{\alpha}, \qquad \alpha, \beta > 0.$$

1. α: elasticity of the index with respect to the $(1 + \text{policy count})$. Higher α increases the relative importance of having many policies.

2. β: elasticity with respect to the average policy score. Larger β rewards higher–quality policies.

3. Often $\alpha + \beta = 1$ is imposed for ease of interpretation, but this is not strictly required.

2.3 Preliminary (Un-transformed) Check of CPSI's Predictive Value

Figure 1 shows how the Climate Policy Stringency Index (CPSI) relates to annual total emissions. For the group of advanced economies, higher index values are *loosely* associated with lower emissions, whereas in many emerging economies the relationship appears to run in the opposite direction.

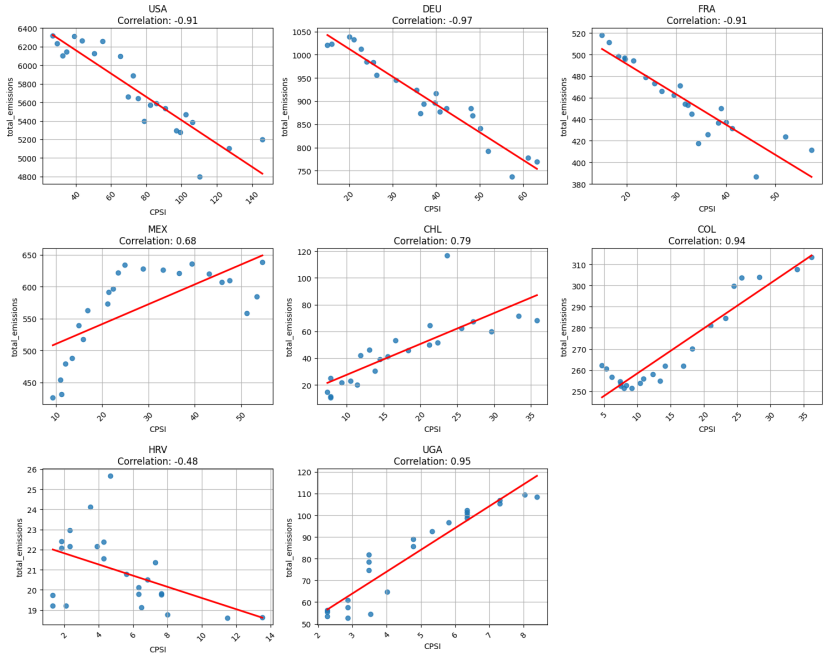

Figure 1: Scatter-plot snapshots of CPSI versus total emissions (selected countries, raw data).